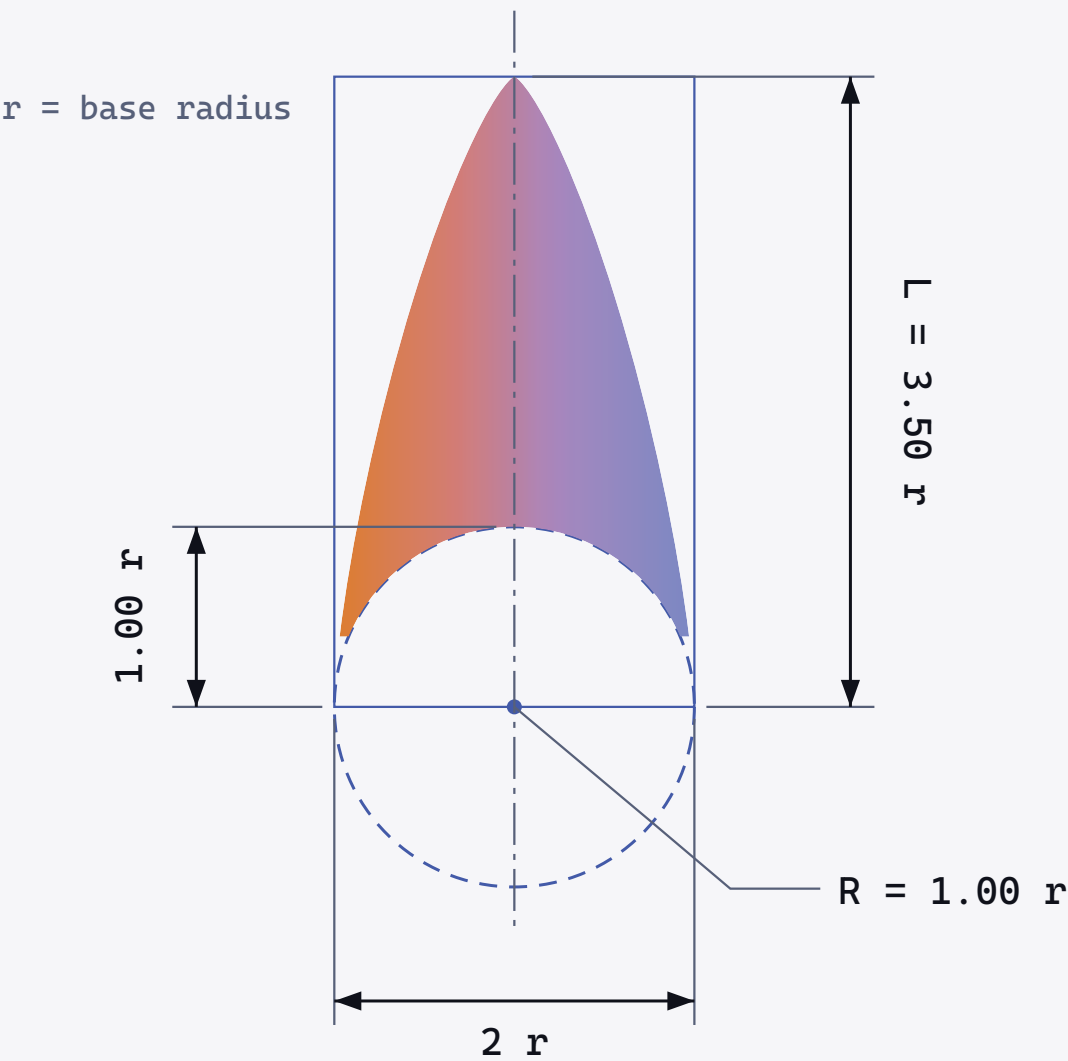
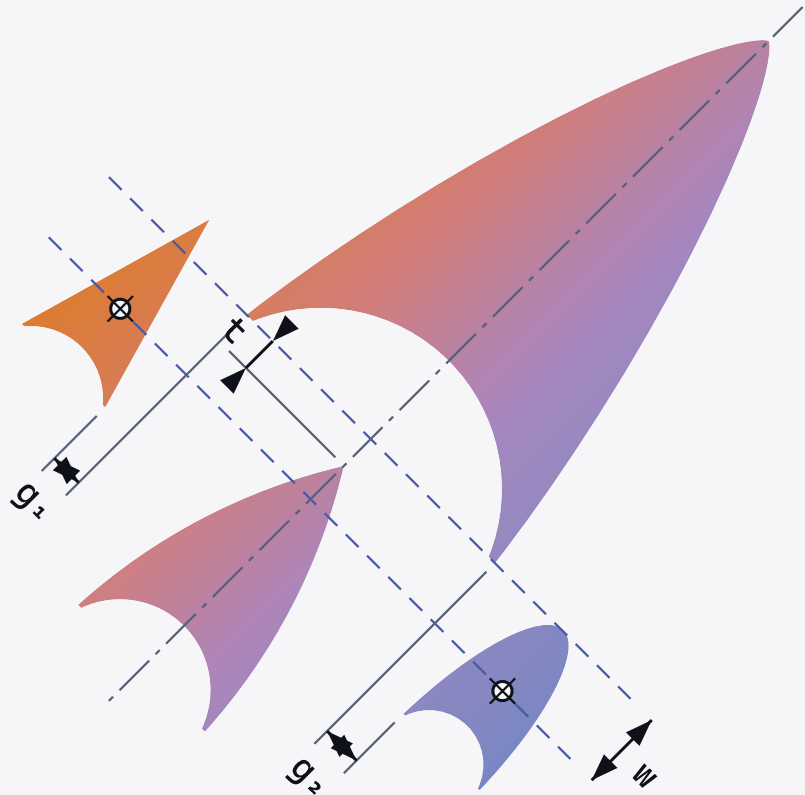


1 · CONE (VON KÁRMÁN)

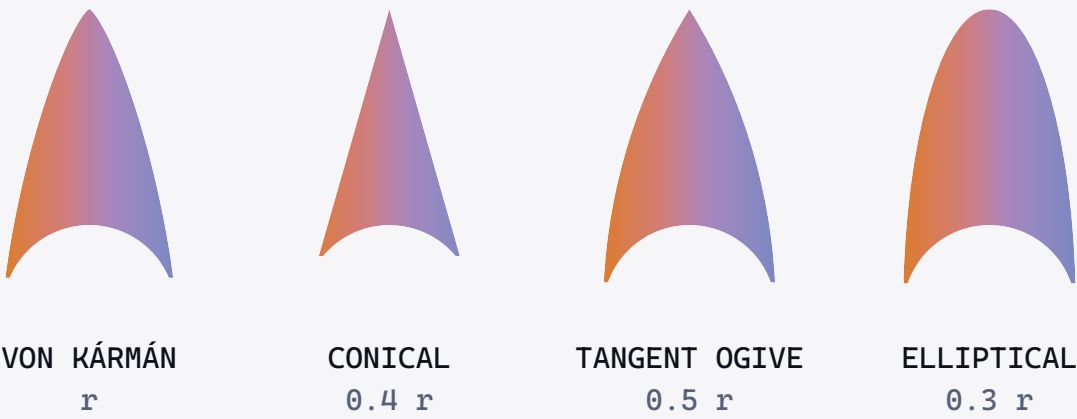


2 · SPACING ON THE 45° GRID

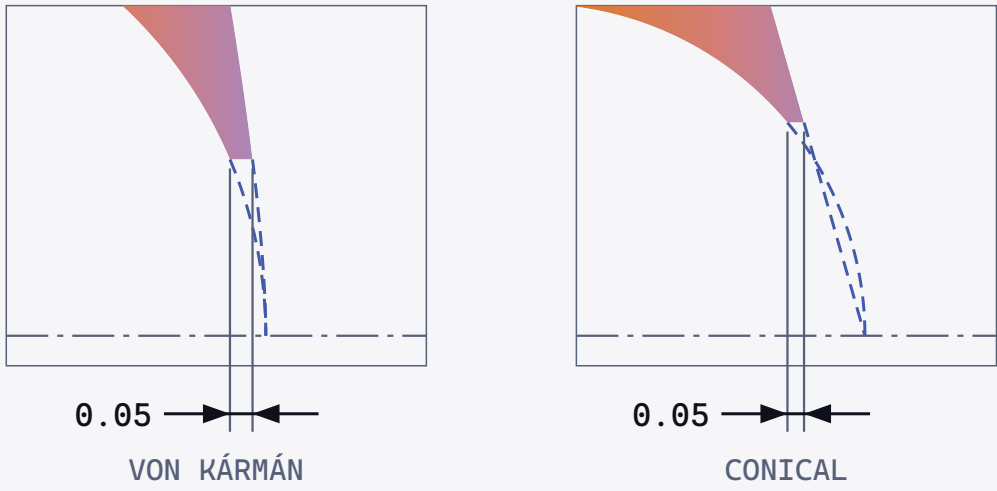


$g_1 = 0.19 r$, $g_2 = 0.23 r$ from the Von Kármán foot corners to the wings. Wing centroids share a line $w = 0.47 r$ below the Von Kármán foot line. The ogive is the tail, on the same axis, tip $t = 0.21 r$ below the Von Kármán foot line. Lay out on the grid, then the grid leans as one piece.

4 · PROFILES (EQUAL L / R)

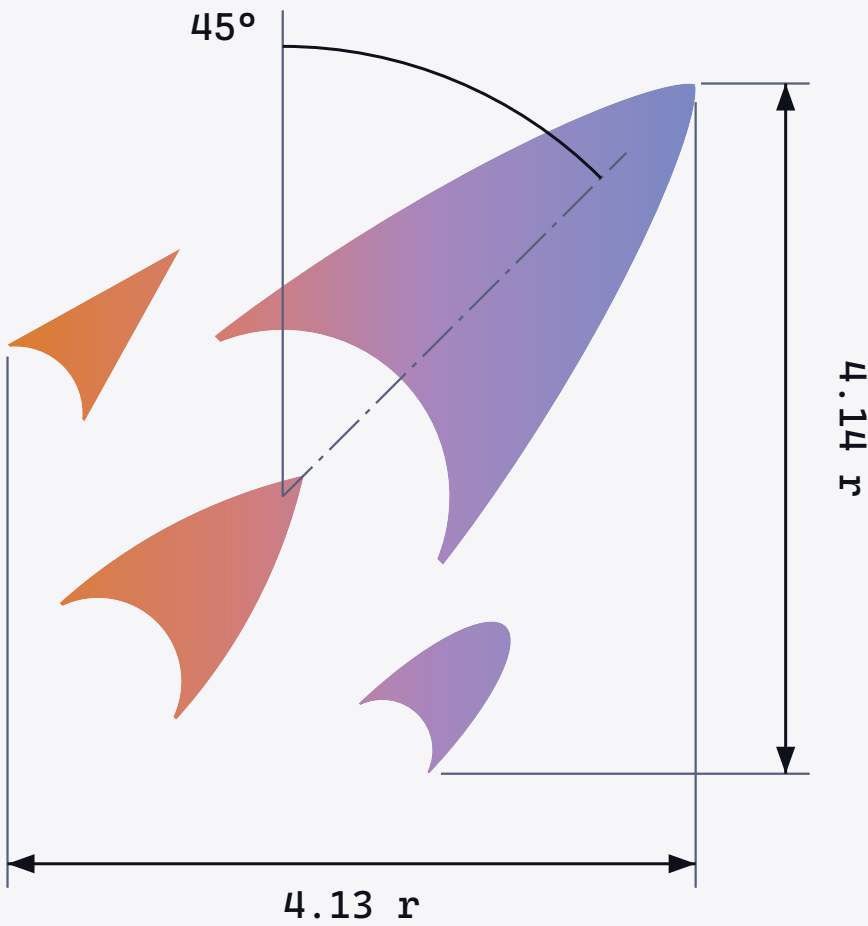


5 · FEET (DETAIL)



Each foot is cut flat, parallel to the base (dash-dot), where the wall between flank and notch is $0.05 \times$ the cone's base radius wide. Dashed: the uncut outline. On the conical cone the notch arc runs outside the straight flank near the base, so the cut also removes that sliver.

3 · CLUSTER



CONE	PROFILE	RADIUS	BASE CENTRE (u, v)
MAIN	VON KÁRMÁN	r	(+0.0, +0.0)
WEST	CONICAL	$0.4 r$	(-1.5, +0.8)
EAST	ELLIPTICAL	$0.3 r$	(+1.5, +0.7)
SOUTH	TANGENT OGIVE	$0.5 r$	(+0.0, +1.6)

u, v in r on the leaned grid, v down.

Standard nose-cone profiles, $L = 3.50 \times$ base radius, notch 1.00 deep, feet cut flat where the wall is 0.05 wide (all $\times$ base radius). Von Kármán is the LD-Haack series ($C = 0$), drawn as smooth curves. The whole cluster leans 45° as one piece. One gradient spans the whole cluster, M orange left to 0 blue right.